

RaceDay API Endpoint Plan

Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new user account as either an Organiser or a Participant.	None (public)	`{ firstName, lastName, email, password, role, phoneNumber }`	201 Created – returns the new user's ID and role. 400 Bad Request – validation failed. 409 Conflict – email already registered.
POST	/api/auth/login	Authenticates a user and returns a JWT token for use on subsequent requests.	None (public)	`{ email, password }`	200 OK – returns JWT token, UserID, and Role. 401 Unauthorized – invalid credentials.

User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Returns the profile of the currently logged-in user.	Any (logged in)	None	200 OK – returns user details (excluding password hash). 401 Unauthorized.
PUT	/api/users/me	Updates the currently logged-in user's own profile details.	Any (logged in)	`{ firstName, lastName, phoneNumber }`	200 OK – returns updated profile. 400 Bad Request – validation failed.

Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Returns a list of all events.	None (public)	None	200 OK – returns array of events.
GET	/api/events/{id}	Returns full details for a single event.	None (public)	None	200 OK – returns event details. 404 Not Found.
POST	/api/events	Creates a new event.	Organiser	`{ eventName, description, eventDate, startTime, venue, city, province }`	201 Created – returns new event with EventID. 400 Bad Request.
PUT	/api/events/{id}	Updates an existing event's details.	Organiser (owner only)	`{ eventName, description, eventDate, startTime, venue, city, province, status }`	200 OK – returns updated event. 403 Forbidden. 404 Not Found.
DELETE	/api/events/{id}	Deletes an event and its associated categories, routes, and weather records.	Organiser (owner only)	None	204 No Content. 403 Forbidden. 404 Not Found.

Categories (master list)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/categories	Returns the master list of category types.	None (public)	None	200 OK – returns array of categories.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/categories	Creates a new reusable category type.	Organiser	`{ categoryName, distanceKm, activityType, description }`	201 Created – returns new category with CategoryID. 400 Bad Request.
PUT	/api/categories/{id}	Updates a category type's details.	Organiser	`{ categoryName, distanceKm, activityType, description }`	200 OK – returns updated category. 404 Not Found.
DELETE	/api/categories/{id}	Deletes a category type.	Organiser	None	204 No Content. 409 Conflict – category is in use.

Event Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Returns all category offerings for an event, with fee and capacity.	None (public)	None	200 OK – returns array of event categories. 404 Not Found.
POST	/api/events/{eventId}/categories	Attaches a master category to this event with a fee and capacity.	Organiser (owner only)	`{ categoryId, entryFee, maximumParticipants }`	201 Created – returns new EventCategoryID. 403 Forbidden. 404 Not Found.
PUT	/api/event-categories/{id}	Updates the fee or capacity for an event's category offering.	Organiser (owner only)	`{ entryFee, maximumParticipants }`	200 OK – returns updated event category. 403 Forbidden. 404 Not Found.
DELETE	/api/event-categories/{id}	Removes a category offering from an event.	Organiser (owner only)	None	204 No Content. 403 Forbidden. 409 Conflict – enrollments exist.

Enrollments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/event-categories/{eventCategoryId}/enrollments	Enrolls the logged-in participant into a specific event category.	Participant	None	201 Created – returns EnrollmentID and status. 404 Not Found. 409 Conflict – already enrolled, or category full.
GET	/api/users/me/enrollments	Returns the logged-in participant's own enrollments and history.	Participant	None	200 OK – returns array of enrollments.
GET	/api/events/{eventId}/enrollments	Returns all enrollments for an event.	Organiser (owner only)	None	200 OK – returns array of enrollments. 403 Forbidden.
DELETE	/api/enrollments/{id}	Cancels the logged-in participant's own enrollment.	Participant (owner only)	None	204 No Content. 403 Forbidden. 404 Not Found.

User Events (registration)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me/events	Returns all events the logged-in participant is registered for, with race number and status.	Participant	None	200 OK – returns array of registrations.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/registrations	Returns all registered participants for an event.	Organiser (owner only)	None	200 OK – returns array of registrations. 403 Forbidden.
PUT	/api/user-events/{id}	Assigns or updates a participant's race number and registration status.	Organiser (owner only)	`{ raceNumber, status }`	200 OK – returns updated registration. 403 Forbidden. 404 Not Found.

Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrollments/{enrollmentId}/results	Captures a finish result for a specific enrollment.	Organiser (owner only)	`{ finishTime, position, averagePace, resultStatus }`	201 Created – returns new result. 403 Forbidden. 404 Not Found. 409 Conflict – result already captured.
GET	/api/event-categories/{eventCategoryId}/results	Returns the full results list for an event category.	None (public)	None	200 OK – returns array of results ordered by position. 404 Not Found.
GET	/api/users/me/results	Returns the logged-in participant's own personal performance history.	Participant	None	200 OK – returns array of the participant's results.

Routes

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/routes	Returns all routes for an event.	None (public)	None	200 OK – returns array of routes. 404 Not Found.
POST	/api/events/{eventId}/routes	Adds a new route to an event.	Organiser (owner only)	`{ routeName, distanceKm, elevationGain, description, mapUrl }`	201 Created – returns new route with RouteID. 403 Forbidden.
PUT	/api/routes/{id}	Updates an existing route's details.	Organiser (owner only)	`{ routeName, distanceKm, elevationGain, description, mapUrl }`	200 OK – returns updated route. 403 Forbidden. 404 Not Found.
DELETE	/api/routes/{id}	Deletes a route from an event.	Organiser (owner only)	None	204 No Content. 403 Forbidden. 404 Not Found.

Weather Information

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/weather	Returns the latest recorded weather information for an event.	None (public)	None	200 OK – returns array of weather records. 404 Not Found.
POST	/api/events/{eventId}/weather	Records a new weather reading for an event.	Organiser (owner only)	`{ temperature, weatherCondition, windSpeed, humidity }`	201 Created – returns new weather record. 403 Forbidden.